David Gómez García

27/10/2026

Electromagnetismo I

$\sigma_P = \bar{P} \underbrace{\hat{n}}_{-\hat{u}_r} = -\frac{Q}{4\pi a^2}\frac{\varepsilon-\varepsilon_0}{\varepsilon}$

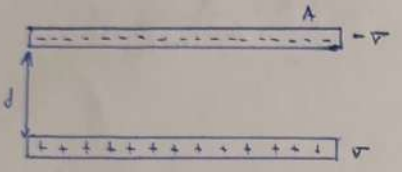
$\oint \bar{E}\cdot d\bar{s} = \frac{1}{\varepsilon_0}(q_{f\,enc} + q_{p\,enc}) = \frac{1}{\varepsilon_0}\cdot\left(Q - \frac{Q}{4\pi a^2}\frac{\varepsilon-\varepsilon_0}{\varepsilon}\cdot 4\pi a^2\right)$

$= \frac{Q}{\varepsilon_0}\left(1-\frac{\varepsilon-\varepsilon_0}{\varepsilon}\right) = \frac{Q}{\varepsilon_0\varepsilon}(\varepsilon-\varepsilon+\varepsilon_0) = \frac{Q}{\varepsilon}$

$E\cdot 4\pi r^2 = \frac{Q}{\varepsilon} \Rightarrow \bar{E}(\bar{r}) = \frac{Q}{4\pi r^2\varepsilon}\hat{u}_r$

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Problema 3 - Hoja de ejercicios tema 4



$C_0 = \frac{Q}{\varphi_+ - \varphi_-} = \frac{\sigma\cdot A}{\frac{\sigma}{\varepsilon_0}\cdot d} = \varepsilon_0\frac{A}{d}$ (capacidad en el vacío)



$\oint \bar{D}\,d\bar{S} = q_{f\,enc}$

$D\cdot S' = \sigma\cdot S' \Rightarrow \bar{D} = \sigma\hat{u}_z$

$-D\cdot S = -\sigma\cdot S \Rightarrow \bar{D} = \sigma\hat{u}_z$

$\left.\begin{matrix}\\\\\end{matrix}\right\} \bar{E} = \frac{\bar{D}}{\varepsilon} = \frac{\sigma}{\varepsilon}\hat{u}_z$

$C = \frac{\sigma\cdot A}{\frac{\sigma}{\varepsilon}\cdot d} = \varepsilon\cdot\frac{A}{d} \Rightarrow \frac{C}{C_0} = \frac{\varepsilon}{\varepsilon_0} = \varepsilon_r = K$ (permitividad relativa)

$\bar{P} = \bar{D} - \varepsilon_0\bar{E} = (\varepsilon-\varepsilon_0)\bar{E} = \frac{\varepsilon-\varepsilon_0}{\varepsilon}\sigma\hat{u}_z$

$\sigma_P = \pm P = \pm\frac{\varepsilon-\varepsilon_0}{\varepsilon}\sigma$

$\oint \bar{E}\cdot d\bar{S}' = E\cdot\bar{S}' = \frac{1}{\varepsilon_0}(q_{f\,enc} + q_{p\,enc}) = \frac{1}{\varepsilon_0}\left(\sigma - \sigma\frac{\varepsilon-\varepsilon_0}{\varepsilon}\right) = \frac{\sigma}{\varepsilon_0}\left(1-\frac{\varepsilon-\varepsilon_0}{\varepsilon}\right) = \frac{\sigma}{\varepsilon_0}\frac{\varepsilon-(\varepsilon-\varepsilon_0)}{\varepsilon} = \frac{\sigma}{\varepsilon}$

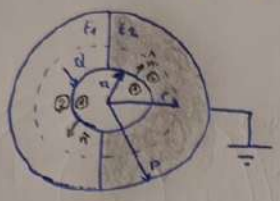
$\bar{E} = \frac{Q}{4\pi\varepsilon r^2}\hat{u}_r \qquad \bar{E} = \frac{\sigma}{\varepsilon}\hat{u}_z$

$\begin{cases}\bar{\nabla}\cdot\bar{E} = \frac{\rho_f}{\varepsilon} \\ \bar{\nabla}\times\bar{E} = 0\end{cases}$

$\bar{E}(\bar{r}) = \frac{1}{4\pi\varepsilon}\int_V \frac{\rho(\bar{r}')\bar{R}}{R^3}dV'$

¡¡Solo si hay un solo medio!!
¡¡No es cierto en general!!

Ejercicio 8 - Hoja de ejercicios Tema 4



$D_{2n} - D_{1n} = 0$

$\bar{E}_{2t} - \bar{E}_{1t} = 0$

$E_1 = E_2 = E$

$D_2 = \varepsilon_2 E \qquad D_1 = \varepsilon_1 \bar{E}$

$\oint \bar{D}\, d\bar{S} = q_{enc}$

$2\pi r^2 \varepsilon_1 E + 2\pi r^2 \varepsilon_2 E = Q \Rightarrow 2\pi r^2 E(\varepsilon_1 + \varepsilon_2) = Q \Rightarrow \bar{E} = \frac{Q}{2\pi r^2(\varepsilon_1+\varepsilon_2)} \hat{u}_r$

$\varphi(r) - \varphi(b) = -\int_b^r \bar{E}\, d\bar{r} = \frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)} \left(\frac{1}{r} - \frac{1}{b}\right)$

$C = \frac{Q}{\varphi(a) - \varphi(b)} = \frac{Q}{\frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)}\left(\frac{1}{a} - \frac{1}{b}\right)} = \frac{2\pi(\varepsilon_1+\varepsilon_2)\,ab}{b-a} = 2\pi(\varepsilon_1+\varepsilon_2)\frac{ab}{b-a} = 2\pi\varepsilon_1 \frac{ab}{b-a} + 2\pi\varepsilon_2 \frac{ab}{b-a} = C_1 + C_2$

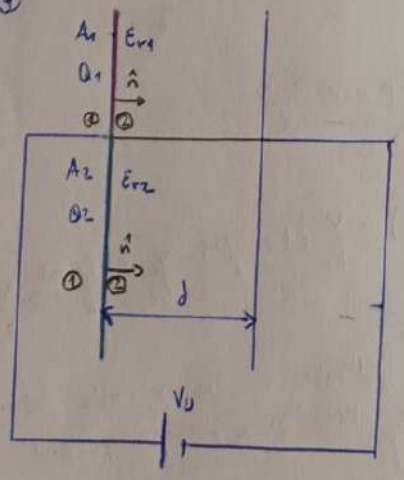
Aplicando las condiciones de contorno

$D_{2n} - D_{1n} = \sigma_f \qquad \bar{E}_{2t} - \bar{E}_{1t} = 0$

$D_1 - 0 = \sigma_1 \;;\; P_1 = \sigma_1 = \varepsilon_1 \cdot E = \frac{Q}{2\pi a^2(\varepsilon_1+\varepsilon_2)} \varepsilon_1$ (Lado izquierdo)

$D_2 - 0 = \sigma_2 \;;\; D_2 = \sigma_2 = \varepsilon_2 \cdot E = \frac{Q}{2\pi a^2(\varepsilon_1+\varepsilon_2)} \varepsilon_2$

④



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Los campos son tangenciales respecto a la superficie de separación de los dos materiales

$E_1 = E_2 = E = \frac{V_0}{d}$

Aplicando las condiciones de contorno

$\sigma_1 = D_1 - 0 \;;\; \sigma_1 = P_1 = \varepsilon_{r1} \cdot E = \varepsilon_{r1} \cdot \frac{V_0}{d}$

Aplicando condiciones de contorno:

$\bar{D} = \sigma = \frac{P\,a}{a+b}$ (placa inferior)

$\sigma = -\frac{P\,a}{a+b}$ (placa superior)

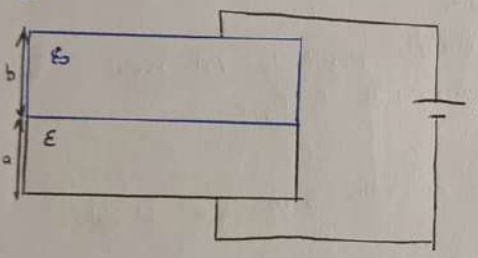
($E = \frac{\sigma}{2\varepsilon_0}$ = campo de una placa)

$\bar{E}_{ext} = P\cdot\frac{a}{a+b}\cdot\frac{1}{2\varepsilon_0} + P\cdot\frac{a}{a+b}\cdot\frac{1}{2\varepsilon_0} = \frac{P}{\varepsilon_0}\frac{a}{a+b}$

(la contribución de las dos placas se anula en el interior del electrete)

$\bar{E}_{int} = -\frac{P}{2\varepsilon_0} - \frac{P}{2\varepsilon_0} + \frac{P}{2\varepsilon_0}\cdot\frac{a}{a+b} + \frac{P}{2\varepsilon_0}\cdot\frac{a}{a+b} = -\frac{P}{\varepsilon_0} + \frac{P}{\varepsilon_0}\left(\frac{a}{a+b}\right) = \frac{P}{\varepsilon_0}\left(1+\frac{a}{a+b}\right) = -\frac{P}{\varepsilon_0}\frac{b}{a+b}$

(11)



$D_{ext} = D_{int} = D$

$\varepsilon_0 E_{ext} = \varepsilon E_{int}$ } En la superficie de separación

$V_0 = E_{int}\cdot a + E_{ext}\cdot b$

$E_{ext} = \frac{\varepsilon}{\varepsilon_0} E_{int}$

$V_0 = E_{int}\left(a + \frac{\varepsilon}{\varepsilon_0} b\right) = \frac{E_{int}}{\varepsilon_0}(\varepsilon_0 a + \varepsilon b)$; $E_{int} = V_0\cdot\frac{\varepsilon_0}{\varepsilon_0 a + \varepsilon b}$

$E_{ext} = V_0\cdot\frac{\varepsilon}{\varepsilon_0 a + \varepsilon b}$

$D = \varepsilon_0 E_{ext} = V_0\cdot\frac{\varepsilon\cdot\varepsilon_0}{\varepsilon_0 a + \varepsilon_0 b}$

$C = \frac{Q}{V_0} = A\cdot\frac{\varepsilon\cdot\varepsilon_0}{a\varepsilon_0 + b\varepsilon}$ ✓

$C = \frac{C_1\cdot C_2}{C_1 + C_2}$ $\quad C_1 = \varepsilon_0\cdot\frac{A}{b}$ $\quad C_2 = \varepsilon\cdot\frac{A}{a}$

$C = \frac{A^2\cdot\frac{\varepsilon_0\,\varepsilon}{ab}}{\left(\varepsilon_0\frac{a}{ab} + \varepsilon\frac{b}{ab}\right)A} = \frac{A\cdot\frac{\varepsilon_0\,\varepsilon}{ab}}{\frac{1}{ab}(\varepsilon_0 a + \varepsilon b)} = \frac{A\cdot\varepsilon\cdot\varepsilon_0}{a\varepsilon_0 + b\varepsilon}$ ✓

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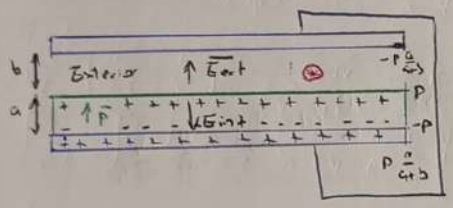
Semana 8a Electromagnetismo

(9)

$\sigma_2 = D_2 - 0$; $\sigma_2 = D_2 = \varepsilon_{r2} \cdot E = \varepsilon_{r2} \cdot \frac{V_0}{d}$

$Q_1 = \sigma_1 \cdot A_1 = \frac{\varepsilon_1 \cdot A_1 \cdot V_0}{d}$; $Q_2 = \sigma_2 \cdot A_2 = \frac{\varepsilon_{r2} \cdot A_2}{d} V_0$

(10)



Tenemos una polarización uniforme e independiente de cualquier campo externo.

El electrete presenta la distribución de cargas de la figura

Esto originará una redistribución de carga en ambas placas, que al estar conectadas, sus potenciales serán el mismo

Al no haber carga libre, el campo D es el mismo en el exterior y en el electrete. ($D_{2n} - D_{1n} = \sigma_f = 0$)

Luego, el E no es el mismo ($\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$)

$\vec{E}_{ext} = \frac{\vec{D}}{\varepsilon_0}$ $\vec{E}_{int} = \frac{1}{\varepsilon_0}(\vec{D} - \vec{P})$

$\varphi_+ - \varphi_- = 0 = \vec{E}_{ext} \cdot b + \vec{E}_{int} \cdot a = 0$

$\varepsilon_0 E_{ext} = \varepsilon_0 E_{int} + P$

$E_{ext} = -\frac{a}{b} E_{int}$

$\varepsilon_0 \left(-\frac{a}{b}\right) E_{int} = \varepsilon_0 E_{int} + P$; $\varepsilon_0 E_{int} \left(1 + \frac{a}{b}\right) = -P$; $E_{int} = -\frac{P}{\varepsilon_0} \cdot \frac{b}{a+b}$

$\vec{E}_{int} = -\frac{\vec{P}}{\varepsilon_0} \cdot \frac{b}{a+b}$

$\vec{E}_{ext} = \frac{\vec{P}}{\varepsilon_0} \cdot \frac{a}{a+b}$

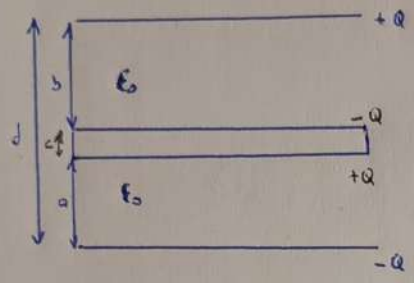
$\vec{D} = \vec{P} \cdot \frac{a}{a+b}$

David Gómez Geraá

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Seminarios Electromagnetismo I

(12)



$$C' = \frac{Q}{\Delta\varphi} = \frac{\sigma \cdot A}{\frac{\sigma}{\varepsilon_0}(a+b)} = \frac{A \cdot \varepsilon_0}{a+b} = \varepsilon_0 \cdot \frac{A}{(d-c)}$$

$$\Delta\varphi = \frac{\sigma}{\varepsilon_0} \cdot a + 0 \cdot c + \frac{\sigma}{\varepsilon_0} \cdot b = \frac{\sigma}{\varepsilon_0}(a+b).$$